KELVIN SHIP WAKE IN THE WIND WAVES FIELD AND ON THE FINITE SEA DEPTH

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ABSTRACT

A kinematics model of the ship wake in the presence of surface waves, generated by wind is presented. It was found that the stationary wave structure behind the ship covered a wedge region with the 16.9° half an angle at the top of the wake and only divergent waves are present in a ship wake for co propagating wind waves. Wind waves field directed at some nonzero angle to the ship motion can cause essential asymmetry of the wake and compressing of its windward half. The extension of Whitham-Lighthill kinematics theory of ship wake for the intermediate sea depth is also presented. The ship wake structure essentially depends from the Froude (*Fr*) number based on the value of the sea depth and ship velocity. For Froude number less than unit both longitudinal and cross waves are presented in the wake region and Kelvin wake angle increased with *Fr*. For $Fr > 1$ wake angle decreased with Froude number and finally only divergent waves are presented in the very narrow ship wake.

Keywords : Kelvin ship wake, Wake angle, Wind waves, Sea depth, Froude number.

1. INTRODUCTION

A moving ship (or any other object moving at or near the water surface) generates a kind of trace on the water surface, which is called a wake. Around and directly behind the ship, the wake is rather complex, with so-called bow and stern waves, eddies, currents, and foam. It depends on the actual shape of the ship, the propeller, and the ship speed, among other factors. From about three ship lengths behind the ship, the main features of the ship wake are rather universal and do not depend much on the shape of the ship or the propeller. Here, the ship wake is a combination of two different phenomena:

1. The turbulent wake, *i.e.*, foam, turbulent water and sometimes surface films in the ship's track.
2. The Kelvin wake, *i.e.* a characteristic wave pattern behind the ship (named after Lord Kelvin who first explained the physics of ship wakes in 1887 [1], Lighthill and Whitham [2] studied the kinematics of ship wake phenomena).

The wave pattern of the Kelvin wake consists of two kinds of waves: Transverse waves (crests across the ship's track) and divergent waves (crests roughly parallel to the ship's track, moving outward). They are confined to a wedge-shaped region behind the ship, and the half angle of that wedge is 19.5°, independent of the ship's speed, as long as the deep water condition is satisfied.

For a typical ship speed of 10m/s (20knots), "deep" as defined above means a depth of more than 5.6m. Since the water depth along shipping lanes in the sea is usually 10m or more, the water can usually be considered deep for most ships except speedboats. The length of the longest waves in the Kelvin wake is between about 15m and 60m for typical ship speeds between 5m/s and 10m/s. The wave's structure behind the ship is presented in Fig. 1.

The two regions at either side of the ship's track where both waves interfere are called Kelvin arms. In these regions the divergent waves gain their largest amplitude and here they are called cusp waves. Often the cusp wave region is defined as the region where the wake half-angle lies between 16.0° and 19.5°.

Comparison of such theoretical results with experiments has shown that, though there is generally good qualitative agreement, there are significant differences between the calculated and measured geometry of the Kelvin ship-wake pattern. One of the striking features of surface ship wakes is the apparent variability of the wake angles [3-6]. Kelvin wake manifestations can be narrower and cover the entire range of half-angles between 9° and the classical deep water 19.5° full Kelvin-wave system half angle [6]. The analysis of about 400 ship wakes in SAR images [7] shows that Kelvin angle generally are narrower than predicted. The wake half-angles in the images are much smaller than 20° in about 20% of cases. One or both Kelvin arms are seen in

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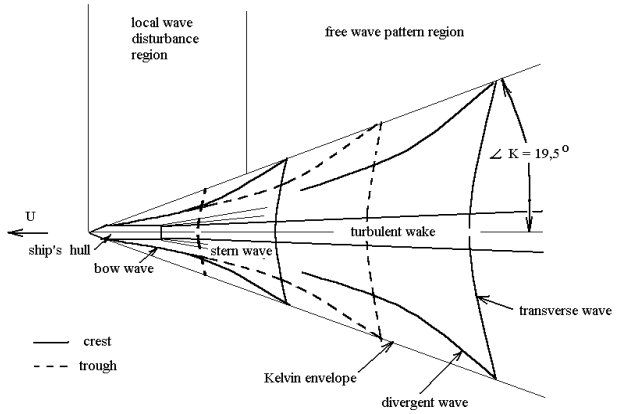


Fig. 1 Scheme of ship wake

other radar images and have the angles close to the classical value for another 80% cases. The radar signatures of Kelvin arms are strongest at low wind speeds and are sensitive to the wind direction. SAR ship wake images presented in [8] for the conditions of wind (~ 2m/s) directed closely to the ship heading gives the trace half-angle ~15°. On the contrary, for the counter propagation of the ship and wind waves the angle of the wake of surface gravity waves can exceed the classical Kelvin wake half-angle [9,10]. Under extremely high wind conditions, no far-wake region can be observed by any remote distance method. Another particularity is connected with asymmetry of observing form of ship wake — very often only one of Kelvin arms is recognized in the SAR images [7].

Research interest to the structure of ship wakes greatly increased in the last 20 years due to using SAR technique for the detection of ships. Radar image of the wake can provide substantial information about the ship, such as its size, direction, and speed of movement [11,12]. Such radar signatures can be used to detect ships even in cases where the ships are not directly visible on the SAR images. This is the case when a ship is too small to be resolved by the SAR or when the sea state is so high that the radar signature of the ship is buried in the speckle of the ambient sea. The estimation of velocity assists the tracking of vessels, which is an important component of ocean surveillance. Recently several image processing techniques have been developed for automatically detecting ship wakes on SAR images [11]. The ship heading and its speed can be estimated from the opening angle of the wake or from the wavelength of the transverse waves in the Kelvin wake [12]. Therefore quantitative estimation of the Kelvin arms geometry is one of the practically important problems.

Discrepancy of the Kelvin solution and observations of the ship wakes may be caused by several reasons, which are followed from the basic assumptions of the model. Kelvin modeled the ship as a prescribed localized pressure distribution and calculated the far-field region, steady state wave disturbance using the method of stationary phase. Near-field behind the ship can be estimated by the length scale $L \sim U^2 / g$, where U is the ship velocity and g is the gravitational acceleration. Far-field correspondingly is placed at the distances $L \sim U^2 / g$ behind the ship. Consideration of the ship like a moving point in a linear water wave theory greatly simplified the problem and neglected beside nonlinearity all effects of the hull form, interference of bow and stern waves, so that is not expected to model accurately the flow field near the ship. Stationary model cannot describe unsteady wakes which, for example can be produced by breaking waves near the ship, and often presented in observations [10].

One more of possible reasons for such a discrepancy are that classical solution considered the ship wake in calm water. But the structure of the ship waves in the presence of external surface wave field can be strongly different from the classical one and possess by some conceptually new properties. To interact visibly in a far-field region ship and wind waves have to be of the same order of magnitude. An intensity of ship waves and the strength of a ship source can be characterized by the dimensionless small parameter $\varepsilon = ag / U^2$, where a is a typical ship wave amplitude. Ship waves will have the scale $O(a\varepsilon^2)$ [13] in the far-field region (at distances $O(\varepsilon^{-6}U^2 / g)$ behind the ship) due to linear dispersion and spreading, consequently amplitudes of wind waves for an effective influencing on the wake region are expected to be of the same order.

The goal of this work is to analyze an influence of wind waves on a ship wake and to suggest a uniformly valid wave model. Directional distribution of surface waves inside the wake will be considered on the basis of kinematics theory of wave's propagation in the moving non-stationary media [2]. One more problem will be analyzed in this paper — the ship wake structure for the finite sea depth. Such a model will extend the results of Whitham-Lighthill [2] ship wake model for the infinite water depth. Detailed analysis of ship waves will permit to determine the structure of subsurface currents behind the ship, its influence on bubbles motion and cavitations dynamic characteristics.

2. STATEMENT OF THE PROBLEM

We'll consider the ship waves dynamics in the moving frame of reference with the ship – source at its origin (P), and uniform liquid flow with constant velocity U in the right direction along the X axis (see Fig. 2).

The main purpose of the present investigation is to analyze the stationary waves structure behind the ship in the presence of external free surface wind wave's pattern. Kinematics approach for a ship wave's motion in calm water first was suggested by Lighthill and Whitham [2].

Dispersion relation for surface gravity waves includes a Doppler shift in this case and has the form:

$$\omega = \vec{U} \cdot \vec{k} + \sqrt{g\,|\vec{k}\,|}\,, \qquad (1)$$

Stationary picture of surface waves takes place for zero value of frequency $\omega = 0$. One can easily show from (1) that ship velocity projection on the direction of the wave propagation has to be equal to phase velocity of surface waves:

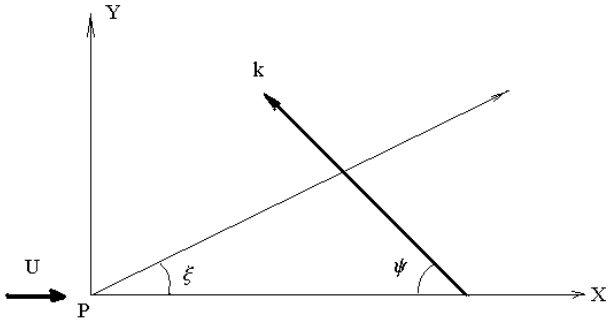


Fig. 2 Geometry of waves behind the ship. ***U*** — flow velocity, ***k*** — wave number vector of surface waves

$$U\cos\psi = \sqrt{g/|\vec{k}|}\,, \tag{2}$$

where (k, ψ) - polar coordinates of wave number vector, see Fig. 2.

Kinematics of wave propagation is governed by the conservation equation for wave number [14]:

$$\frac{\partial\vec{k}}{\partial t} + \nabla\omega = 0\,, \tag{3}$$

and consistency relation for phase function:

$$\frac{\partial k_1}{\partial Y} = \frac{\partial k_2}{\partial X}\,. \tag{4}$$

Equation (3) is identically valid for the stationary problem and zero frequency of surface wave field.

So, finally we have to solve pair of Eqs. (2) and (4) for two unknown functions (k_1, k_2). Dependence $k_1 = f(k_2)$ is known from the relation (2) and so the quasi- linear Eq. (4) can be analyzed by the method of characteristics. Let us rewrite it as an equation for k_2 function:

$$\partial k_2/\partial X - f'(k_2)\,\partial k_2/\partial Y = 0 \tag{5}$$

We note that the left side of Eq. (5) is the total derivative of k_2 along the characteristic curve with the slope $DY/DX = -f'(k_2)$:

$$\frac{Dk_2}{DX} = \frac{\partial k_2}{\partial X} + \frac{DY}{DX}\frac{\partial k_2}{\partial Y} = \frac{\partial k_2}{\partial X} - f'(k_2)\frac{\partial k_2}{\partial Y} = 0\,,$$
$$\frac{DY}{DX} = -f'(k_2)\,. \tag{6}$$

Since the total derivative of k_2 is zero, the value of k_2 is constant for every characteristic curve and from the second of Eq. (6) follows that they all have to be a straight lines in the (X, Y) coordinates.

For the point source P we have the central wave, characteristics are straight rays with the constant wave number along the ray:

$$Y/X = -f'(k_2)\,;\;\; k_2 = \text{const}\,, \tag{7}$$

By using the relation (1), the function $f'(k_2)$ can be easily calculated:

$$\frac{df}{dk_2} = -\frac{\dfrac{\partial\omega}{\partial k_2}}{\dfrac{\partial\omega}{\partial k_1}}\,, \tag{8}$$

Substituting (2) and (8) into the first of the expressions (6) gives the following equation for the ray directions:

$$tg\xi = \frac{Y}{X} = \frac{-k_2/k_1}{1+2\left(\dfrac{k_2}{k_1}\right)^2}\,. \tag{9}$$

here ξ is the ray direction (see Fig. 2). The phase synchronism condition (2) and the equation for ray direction (9) define the signs of wave-number components $k_1 < 0$, $k_2 > 0$ for $X > 0$, $Y > 0$. Picture of waves is axis-symmetrical with respect to X axis, so we'll only consider the region $X > 0$, $Y > 0$. Function $\xi = \xi(-k_2/k_1)$, defined by the Eq. (9) has the maximum, equal to $\xi_m = 19.5°$. Every crest line has a sharpness point at this critical ray, dividing wave system into longitudinal ($-k_2/k_1 < 1/\sqrt{2}$) and cross ($-k_2/k_1 > 1/\sqrt{2}$) parts. Typical crest curve is presented at Fig. 3, and common wave's scheme in Fig. 1.

Therefore main results of Kelvin solution for the ship wake are the following:

- Wave number vector function is constant at every ray starting from the source point P (Fig. 2);
- Two systems of waves are presented inside the ship wake: Transverse waves (crests across the ship's track) and divergent waves (crests roughly parallel to the ship's track, moving outward). They are confined to a wedge-shaped region behind the ship, and the half angle of that wedge is 19.5°. The angle geometry of the wake does not depend on the ship speed and determined only by dispersive properties of the surface gravity waves. The wave picture is axis-symmetrical related to the direction of ship motion.

3. ONE-DIMENSIONAL INTERACTION OF SHIP WAKE AND EXTERNAL SURFACE WAVE'S FIELD

We'll begin consideration with the simplest, but very important case of co propagation of the ship and surface wind waves along the X-axis in the phase synchronism conditions:

$$\vec{U} = \sqrt{\frac{g}{|k_{10}|}}\frac{\vec{k}_{10}}{|k_{10}|}\,, \tag{10}$$

here $\vec{k}_{10}$ is the wave-number of the external wave field. Constant wave-number field (k_{10}, $k_{20} = 0$) satisfied both kinematics Eqs. (2) and (5) with zero wave frequency in the chosen moving frame of reference:

$$Uk_{10} + \sqrt{gk_{10}} = 0\,. \tag{11}$$

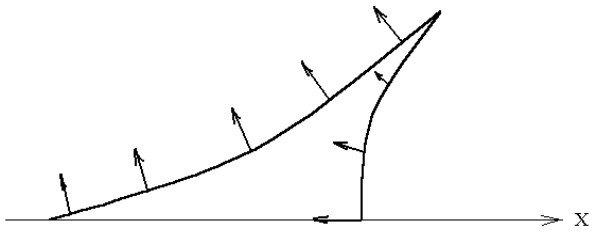


Fig. 3 Crest lines of waves in the ship trace

But continuous solution for wave-number function, including both ship wake region and external field is impossible due to different values of the wave-number vector along the critical rays $\xi = \xi_m$. Weak solution of the problem, including breaks of functions will be constructed here. The usual structure for such kind of solutions has two or more regions of continuous solution, joined by break lines; functions have a jump across these lines, nevertheless the integral physical conservation laws are still valid there and give the connection between values before and after the break line [15]. It is well known, that for quasi-linear differential equation of the first order like the Eq. (5) exists infinite number of conservation laws and special consideration has to be done for making a choice between them. In our case the Eq. (5) has a clear physical sense: It is the consequence condition, necessary for existing the phase of wave motion. Taking in mind the common definition of wave-number function:

$$\vec{k} = \nabla\theta \,,$$

where θ is the phase of wave motion, we'll assume the integral conservation law, expressing the continuity of phase function along the break line with direction *D*:

$$[k_1] + D\,[k_2] = 0 \,, \qquad (12)$$

where square brackets [ ] expressed the jump of function before and after the break: $[k_i] = k_i - k_{i0}$, i = 1, 2. Direction of break line *D* can be expressed from Eqs. (2), (11) and (12) as a pure function of wave-number components ratio $(-k_2 / k_1)$ immediately after the jump:

$$D = -\frac{k_1}{k_2}\left(1 - \frac{1}{\left(1 + (k_2/k_1)^2\right)^{1/2}}\right) \,, \qquad (13)$$

Displacement of the break line *D* is founded by solving the consequence equation, following from (9) and (13):

$$-\frac{k_1}{k_2}\left(1 - \frac{1}{\left(1 + (k_2/k_1)^2\right)^{1/2}}\right) = -\frac{k_2}{k_1}\frac{1}{1 + 2(k_2/k_1)^2} \,, \qquad (14)$$

Graphical solution of Eq. (14) is presented in Fig. 4.

Most significant result here is that the dashed break line, defined by direction *D*:

$$\xi_0 = \operatorname{arctg} D \approx 16.9° \,. \qquad (15)$$

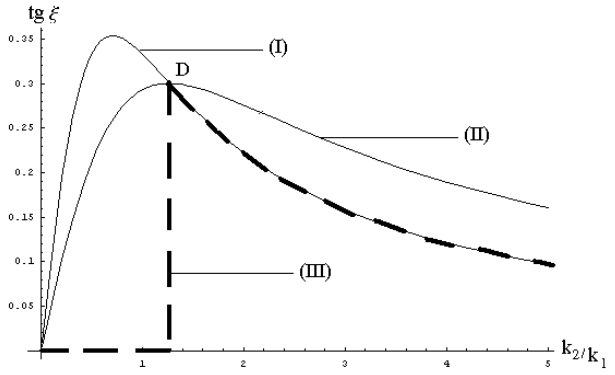


Fig. 4 Graphical solution of the equation (14); Curve (I) represents solution for ship wake in calm water; curve (II) – possible directions of the break line; curve (III) – combined weak solution. Intersection point *D* defines the unique position of the break line

reduces the maximal angle of ship wave's structure (ξ_m = 19.5°). Another which may be the most important particularity of the weak solution is, that it eliminates longitudinal part of ship waves — only divergent waves behind the ship are presented by high frequency part of the solution (see curve (III), Fig. 4) to the right of the intersection point *D*.

Final combined solution (dashed break line (III) at Fig. 4) includes the external surface wave's field of zero frequency (11), breaking line defined by direction ξ_0 and relatively high frequency divergent waves inside the ship wake.

4. TWO-DIMENSIONAL INTERACTION OF SHIP WAKE AND EXTERNAL SURFACE WAVE'S FIELD

Let us consider next the stationary surface waves, which have some nonzero angle of propagation to the axis of ship movement. Observer on the ship will see stationary surface picture of constant wave-number field $\vec{k}_{10} = (k_{10}, k_{20})$ for waves with zero frequency:

$$Uk_{10} + \sqrt{g\,|\vec{k}_{10}|} = 0 \,, \qquad (16)$$

It can be emphasized, that wave's picture can not be axis-symmetrical in this case due to nonzero value of vertical wave number component k_{20}.

Continuous solution for wave-number function, including both ship wake angle region and external field is impossible due to different values of wave-number vector along the critical rays $\xi = \xi_m$ at the boundaries of ship wake. Weak solution of the problem, including breaks of functions will be constructed here. We'll assume the validity of first integral conservation law and express the continuity of phase function along the break line with direction *D*:

$$[k_1] + D[k_2] = 0 \,, \qquad (17)$$

where square brackets [ ] express the jump of function before and after the break: $[k_i] = k_i - k_{i0}$, $i = 1,2$. Direction of break line D can be expressed from Eqs. (16) ~ (17) as a function of external wave-number angle $tg\psi_0 = k_{20} / k_{10}$ and wave number angle immediately after the jump $tg\psi = k_2 / k_1$, *i.e.*

$$D = -\frac{(1+tg\psi^2)^{1/2} - (1+tg\psi_0{}^2)^{1/2}}{tg\psi(1+tg\psi^2)^{1/2} - tg\psi_0(1+tg\psi_0{}^2)^{1/2}}\ , \qquad (18)$$

The displacement of the break line D is founded by solving the consequence equation, expressing the fact that the value of wave-number function at the shock ray (18) must be equal to the corresponding value of continuous solution from Eqs. (7) and (9) at the same direction:

$$-\frac{(1+tg\psi^2)^{1/2} - (1+tg\psi_0{}^2)^{1/2}}{tg\psi(1+tg\psi^2)^{1/2} - tg\psi_0(1+tg\psi_0{}^2)^{1/2}} = -\frac{tg\psi}{1+2tg\psi^2}\ . \qquad (19)$$

For every direction of external wave field $tg\psi_0$, the relation (19) implicitly determines the direction of wave number immediately after the jump $tg\psi$ and correspondingly the possible directions of break line D. The graphical representation of solution of the Eq. (19) is presented in Fig. 5.

Curves (I) and (II) in Fig. 5 represent the possible directions of the strong break lines of wave number function for downward $k_{20} < 0$ and upward $k_{20} > 0$ directions of external wave field, correspondingly. Curve (III) expresses the solution of the Eq. (19): $tg\psi = tg\psi_0$ which evidently indicates the possibility of smooth transition from internal to external wave's field.

Now we'll notice that if $tg\psi$, $tg\psi_0$ and D satisfied Eqs. (16) ~ (19), then $-tg\psi$, $-tg\psi_0$ and $-D$ will also be the solution of these equations. Therefore, for modeling the common scheme of waves behind the ship it is sufficient to consider only the upper semi plate $\xi > 0$ and various directions of external surface wave field $tg\psi_0$. For downward external waves $k_{20} < 0$ strong break line (corresponding to points of the curve (I), Fig. 5) will be placed inside the wake, cut the transverse part of Kelvin waves and provide the transition from divergent waves inside the wake to the external field. The upward external wave field can be smoothly joined with Kelvin solution by using the solution represented at curve (III) in Fig. 5 and both transverse and divergent waves will be presented in the ship wake for such a case. On the basis of described solution, one can easily construct the common scheme of the ship wake in the presence external wave field. A principle scheme of the common solution is presented in Fig. 6.

Interaction of the ship wake with free surface waves under phase synchronism conditions (16) will cut some part of ship waves directed oppositely to the wave's field and correspondingly the angle of this region of ship wake will be narrower than the classical one. "Shadow" half of ship wake roughly will keep it initial structure and still consist of divergent and transverse systems of waves.

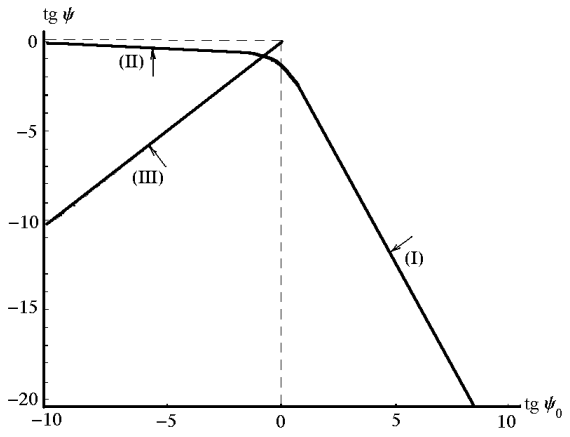


Fig. 5 Graphical representation of function $tg\psi = f(tg\psi_0)$

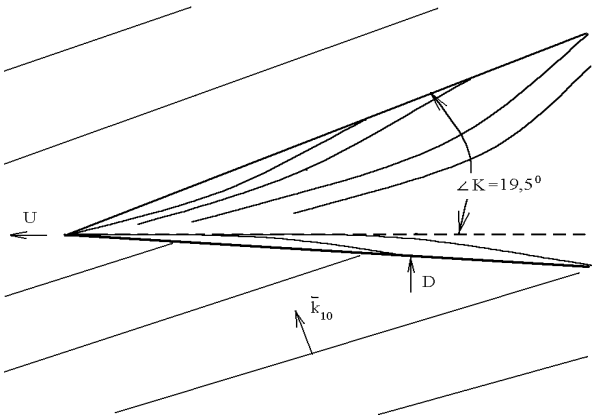


Fig. 6 Structure of ship wake under external surface waves field

5. SHIP WAKE FOR A FINITE DEPTH OF THE SEA MODEL

Finally we'll examine the ship wake structure for the intermediate depth of the sea $H < \infty$ by the methods of kinematics wave theory.

Dispersion relation for surface gravity waves in this case has the form:

$$\omega = \vec{U}\cdot\vec{k} + \sqrt{g\,|\,\vec{k}\,|\tanh(kH)}\ , \qquad (20)$$

Following the same scheme of the ray method as in chapter 2 for the wave consistency Eq. (5) together with zero frequency condition $\omega = 0$ we receive the following expressions for the ray directions with the constant values of wave number components ($k_1 = \text{const}$, $k_2 = \text{const}$) along it:

$$\tan\xi = \frac{Y}{X} = \frac{\sqrt{\dfrac{kH\ Fr^2}{\tanh(k\mathrm{H})} - 1}\ \cdot \left(1 + \dfrac{2kH}{\sinh(2kH)}\right)}{2 + 2\left(\dfrac{kH\ Fr^2}{\tanh(k\mathrm{H})} - 1\right) - \left(1 + \dfrac{2kH}{\sinh(2kH)}\right)}\ ,$$

where $k = |\,\vec{k}\,|$ is the length of wave number and $Fr = U/\sqrt{gH}$ — Froude number for the considering problem.

The ship wake structure strongly depends from the value of Froude number.

For every value of $Fr < 1$ the ship wake still consists from a two waves system of longitudinal and cross waves covered the wedge region with angle at the origin depending from Fr. Dependence of wake angle from the Froude number is presented in Fig. 7 (Curve I). One can see that the wake angle is very closed to the Kelvin value for the low Froude numbers $Fr << 1$ and substantially increased with Froude for $0.5 \le Fr < 1$. Finally half of a wake angle approaches to $\pi/2$ for Froude closed to unit. The direction of waves at the cusp line also vary from the Kelvin values for small $Fr << 1$ to almost collinear with the ship velocity direction for $Fr \sim 1$. Direction of cusp waves in dependence of Fr number is also presented at Fig. 7 (Curve II).

The ship wake structure is drastically modified for $Fr > 1$. The wake angle and direction of cusp waves in dependence from Fr are presented at Fig. 8. Most long waves are placed at the edges of wake and only one system of waves is presented for every ray inside the wake region. Wake angle decreased with Froude number from π for $Fr \sim 1$ to really small values for $Fr >> 1$ and only divergent waves with the crests almost collinear with ship track are finally presented in the very narrow ship wake.

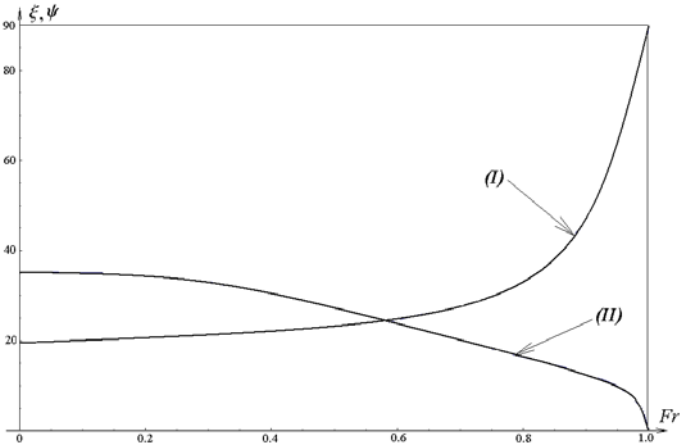


Fig. 7 Curve (I) – ship wake angle, Curve (II) – direction of cusp waves in dependence of Froude ($Fr < 1$) number

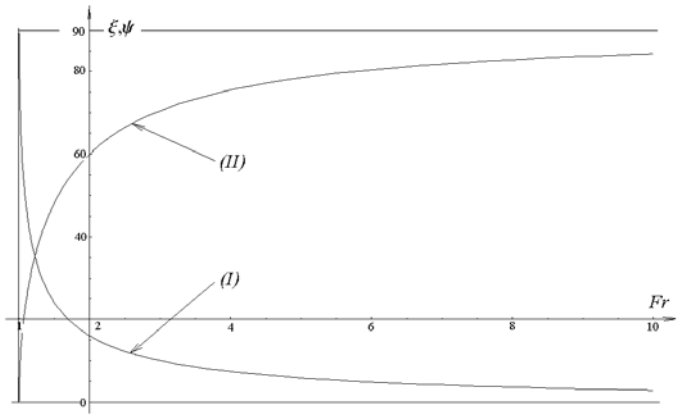


Fig. 8 Curve (I) – ship wake angle, Curve (II) – direction of cusp waves in dependence of Froude ($Fr > 1$) number

6. CONCLUSIONS

A lot of observations on the structure of the Kelvin ship waves in the presence wind waves show its essential difference from the classical one. No stationary wake angle on the counter-propagating wind field can exceed the value of 39°. On the contrary, co propagating ship and wind waves can give the effect of reducing of the angle between Kelvin arms.

The kinematical model presented here describes possible modification for the classical Kelvin ship wake solution in the presence of wind surface waves. Ship wake can be narrower and loses its axis of symmetry.

The Kelvin ship wake for the intermediate water depth strongly depends from the Froude number of the problem. For the $Fr < 1$ the wake angle increased with Fr and finally riches the value of π for Fr equal to unit. Two systems of waves – longitudinal and divergent are still presented in the wake. For $Fr > 1$ the wake angle oppositely decreases with Fr and only one type of waves presented in the wake region. The longest waves are placed at the cusp lines of the wake. Finally for a very big values of $Fr >> 1$ the wake angle has a very little value with the only divergent waves inside it and directed almost normally to the ship track.

Analysis is carried out for linear or weakly nonlinear surface wave field. The above mentioned effects can be visible in the far-field of ship wake for essentially developed wind waves of the same order as divergent ship waves.

But, more precisely, region of applicability for the present model can be estimated by a series of corresponding experiments in laboratorial or natural conditions.

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